

Geragogy

The Key to Learning for the Aging Population — why teaching older adults is a science of its own, and what happens when technology finally takes it seriously

A research brief from mynaani — September 2026

Executive summary

If you are responsible for the learning, engagement, or independence of older adults — in a senior living community, a health plan, or an aging-services organization — you have likely watched a well-intentioned program fail for reasons nobody could name. The content was fine. The learners were willing. The method was wrong.

This paper examines geragogy: the evidence-based study of how adults learn in later life. It is not pedagogy applied downward, and it is not "senior-friendly" styling applied on top. It is a distinct discipline, grounded in decades of cognitive-aging, vision-science, and instructional-design research, organized around one central finding: for older learners, the barrier is rarely ability — it is design that ignores how aging changes perception, attention, working memory, and confidence.

The research record is clear on the mechanism. Contrast sensitivity declines roughly 80% by age 80. The brain's ability to filter out irrelevant visual information weakens with age, so clutter and motion don't just distract — they consume the capacity learning requires. Cognitive load is the dominant mediator of digital-learning outcomes in older adults. And confidence — self-efficacy — is the strongest predictor of whether an older learner persists or quietly withdraws.

This paper lays out what geragogy demands of curriculum, of visual and interface design, and of the software systems that deliver learning — a discipline we call cognitively-protective learning design. It then offers decision-makers a practical evaluation framework, and closes with an honest account of what the evidence does and does not yet prove.

1. The problem: we teach older adults as if nothing has changed

Nearly every educational product an older adult will ever touch was designed for a younger nervous system — and then, at best, "adjusted": larger buttons, slower pacing, simplified vocabulary. The adjustment treats the symptoms. The underlying assumption remains: *the learner adapts to the design*.

The result is measurable. In Nielsen Norman Group's long-running program of senior usability research, adults 65+ succeed at standard website tasks only about 55% of the time, vs. 75% for adults 21–55 — a gap that has persisted across two decades of interface "improvement." Only about one in four internet users 65+ reports feeling very confident performing online tasks (Pew). And 59% of adults 50+ say technology "isn't designed with them in mind" (AARP, 2025).

These are not ability deficits. They are design deficits — and they compound: an interface that overloads a learner produces errors, errors erode confidence, and lost confidence ends the learning. The failure is assigned to the learner when it belongs to the method.

2. What geragogy is — and how it differs from what came before

Geragogy is the theory and practice of teaching older adults: the study of how aging changes the conditions of learning, and of the methods that honor those changes.

- **Pedagogy** — the teaching of children — assumes a learner building competence from zero, under authority.
- **Andragogy** — the teaching of adults — assumes a self-directed learner with experience to build on.
- **Geragogy** adds what andragogy misses: that aging measurably changes perception, attention, memory, and the emotional conditions of learning — and that *confidence preservation* is as important as content delivery (Boulton-Lewis, *Educational Gerontology*; the andragogy lineage traces to Knowles). The older learner brings a lifetime of competence; a method that makes them feel incompetent loses them regardless of content quality.

Geragogy's core commitments — predictability, dignity, reversibility, confidence preservation, containment of cognitive load — are not courtesies. They are the conditions under which learning remains possible at all.

3. The science: what aging actually changes

Decades of peer-reviewed and institutional research converge on four mechanisms:

Vision. By age 80, typical contrast sensitivity is roughly 80% lower than at 20; presbyopia reduces near focus, and pupil shrinkage reduces retinal illumination (W3C/WAI literature review; Owsley, UAB School of Medicine). Small type, low contrast, and dense screens are not "harder to read" for older adults — they are *costlier to process*, consuming attention the lesson needs.

Attention. Aging reduces inhibitory control — the brain's ability to suppress irrelevant information (Hasher & Zacks, University of Toronto; 1988 forward). Decorative imagery,

pop-ups, badges, and motion do not sit politely in the background; they are processed, and they degrade comprehension and memory for what actually matters.

Working memory and load. Cognitive load is the strongest mediator of digital-learning outcomes in older adults — in a 2025 JMIR experiment, load correlated negatively with learning performance at large effect size. Every unnecessary element, unfamiliar icon, or surprising state change spends working memory that was budgeted for the material.

Confidence. Self-efficacy is the strongest predictor of persistence in older learners — and it is trainable: randomized controlled evidence (Laganà et al.) shows that age-appropriate technology training significantly improves attitudes and self-efficacy. Design that protects confidence keeps learners in the room; design that wounds it empties the room quietly.

The corollary is sobering: conventional "more is more" design — rich visuals, animation, gamified dashboards, streaks and badges — is not neutral for older learners. Each of these devices taxes the exact mechanisms age has changed. What reads as engaging to a 30-year-old reads as noise, pressure, or bewilderment to an 80-year-old. Visual richness, deployed carelessly, is an active harm.

4. What geragogy demands: cognitively-protective learning design

If the science is taken seriously, it stops being a style guide and becomes engineering. We call the resulting discipline **cognitively-protective learning design** — geragogy enforced at every layer of the system:

The curriculum layer. Short, structured lessons in plain language. Worked examples before open-ended tasks — the worked-example effect is among the most robust findings in instructional science (Kalyuga). No tests, grades, timers, or streaks: evaluative pressure consumes working memory and signals judgment. Progress that is visible and reversible, never gamified into anxiety.

The visual layer. Readable type as a floor, not a preference; high contrast; hard limits on how much appears on screen at once; layouts that do not move; no decorative motion; iconography only beside words. This is not austerity — it is respect for an attention system that can no longer afford waste.

The system layer. The step the field has largely skipped: enforcement. A design intention decays the moment an interface changes unpredictably. Mynaani's patent-pending cognitively-protective system governs interface behavior itself — transitions, density, and pacing are constrained so no screen can exceed a cognitively safe state for the learner. Geragogy stops being a checklist reviewers hope was followed and becomes a property the software guarantees.

That is the difference the word makes. "Senior learning" adjusts content. Geragogy re-engineers the conditions of learning.

5. What decision-makers should look for

For operators and plans evaluating learning programs for older adults, five tests follow directly from the evidence:

- **Method before content.** Ask *what* in the program is traceable to geragogical research — not whether it says "senior-friendly."
- **Confidence as a measured outcome.** If the program cannot tell you how it protects and builds self-efficacy, it is missing the variable that decides whether learners persist.
- **Load, by design.** Ask how on-screen density, pacing, and state changes are bounded — and who or what enforces the bound.
- **No borrowed pressure.** Tests, timers, streaks, and gamified urgency are evidence of a method built for a different nervous system.
- **Honest evidence.** Programs that cite sources, acknowledge limits, and invite evaluation are safer partners than programs that promise.

What success looks like within a year: learners who approach the next new technology with less apprehension, not more; staff freed from ad-hoc tech support; and families who describe confidence rather than concern.

6. Limitations and open questions

This brief synthesizes third-party research; it is not a report of mynaani's own outcome studies, which do not yet exist. The component findings — contrast sensitivity, inhibitory decline, load mediation, self-efficacy — are each well-established; their combination into a single engineered discipline awaits, and deserves, formal field evaluation in residential settings. "Patent pending" describes an approach under disclosure, not a validated outcome. We publish this brief to be checked, not merely believed.

7. Conclusion and next step

The learning needs of the aging population are not a niche. They are a foreseeable condition of a society whose median age is rising and whose services are moving online. Geragogy is the discipline built for that condition — and programs that adopt it will outlast those that repaint younger design in softer colors.

Mynaani is onboarding a small number of founding communities to put cognitively-protective learning design into practice — and to help build the field evidence this

discipline deserves. To discuss a pilot or request this brief for your team, write to hello@mynaani.com.

About mynaani

Mynaani is an AI learning program designed specifically for adults 55+, built on geragogical principles and a patent-pending cognitively-protective interface. We have an interest in the conclusion of this paper; readers should weigh it accordingly. Every statistic cited is third-party and verifiable below.

References

- Nielsen Norman Group. *Usability for Senior Citizens: Improved, But Still Lacking*. <http://www.nngroup.com/articles/usability-seniors-improvements/>
- Pew Research Center (2017). *Barriers to adoption and attitudes towards tech among older Americans*. <https://www.pewresearch.org/internet/2017/05/17/barriers-to-adoption-and-attitudes-towards-technology/>
- AARP Research (2025). *2025 Tech Trends and Adults 50-Plus*. doi:10.26419-2fres.00891.001
- W3C Web Accessibility Initiative. *Web Accessibility for Older Users: A Literature Review*. <https://www.w3.org/WAI/older-users/literature/>
- Owsley, C. (2016). *Vision and Aging*. Annual Review of Vision Science 2:255–271. UAB School of Medicine. <https://www.annualreviews.org/content/journals/10.1146/annurev-vision-111815-114550>
- Hasher, L. & Zacks, R.T. (1988). *Working memory, comprehension, and aging*. Psychology of Learning and Motivation 22:193–225. University of Toronto. https://hasherlab.psych.utoronto.ca/abstracts/hasher_zacks_88.htm
- Campbell, K.L. et al. (2014). *The disruptive — and beneficial — effects of distraction on older adults' cognitive performance*. Curr. Dir. Psychol. Sci. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3927084/>
- JMIR (2025). *Cognitive Load and Learning Performance in Digital Health Education for Older Patients*. <https://www.jmir.org/2025/1/e79430>
- Laganà, L. et al. *Enhancing computer self-efficacy and attitudes in multi-ethnic older adults: a randomised controlled study*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4265211/>
- Kalyuga, S. — worked-example effect literature (instructional science).

- National Institute on Aging / U.S. National Library of Medicine. *Making Your Web Site Senior Friendly*. <https://repository.arizona.edu/handle/10150/106378>
- AARP Research (2026). *Skills Gap Persists Even as Older Adults Embrace Digital Life*. <https://www.aarp.org/pri/topics/technology/internet-media-devices/digital-literacy-skills/>
- npj Digital Medicine (2026). *Digital exclusion and health in older adults: a systematic review*. <https://www.nature.com/articles/s41746-026-03091-6>
- Pew Research Center (2026). *How Americans' opinions and use of AI differ by age*. <https://www.pewresearch.org/internet/2026/06/17/how-opinions-and-use-of-ai-differ-by-age/>
- Boulton-Lewis, G.M. (2010). *Education and Learning for the Elderly: Why, How, What*. Educational Gerontology 36(3). (Geragogy as a distinct discipline.)
- Knowles, M.S., Holton, E.F. & Swanson, R.A. *The Adult Learner* — the andragogy lineage geragogy extends.

© 2026 mynaani. This document may be shared freely with attribution.